Predicting Arrest Outcomes in Chicago: A Data Mining Approach Using Clustering and Predictive Modeling

MIS 576: Data Mining

Team Name: InsightOps

Team Members

Name	Email ID
Atharva Chaskar	achas6@uis.edu
Ajay Jayshree	ajay10@uis.edu

University of Illinois Springfield

Abstract and Highlights

Urban crime continues to be one of the most serious public safety issues which is faced by major American cities. This project analyzes the crime data from the City of Chicago between 2018 and 2020. This is the period when the whole world was hit by the COVID-19 pandemic. The first goal of the project was to analyze the data and uncover hidden patterns in where and when crimes occur, and the second was to build models that can predict whether a reported crime will lead to an arrest.

The data was retrieved from the Chicago Data Portal and originally included around 9.5 million records. After narrowing it down to the 2018-2020 window and removing the small fraction of records missing location information which is about 2% of the complete data, we had a cleaned dataset of roughly 750,000 records. Due to storage limitations in the UIS Citrix environment, we worked with a 25% sample for modeling purposes. Before any analysis, few steps were performed in Python like parsing dates, converting boolean values, and selecting the most relevant variables.

To explore patterns in the data, we used hierarchical clustering with three different methods (Ward, Centroid, and Average linkage). This showed two to three consistent crime clusters, distinguished by location, time of day, and neighborhood characteristics. We tested the stability of these clusters using a 90% sub-sample, and the results confirmed the stability of the clusters, particularly for the Ward method.

For the predictive analysis, we built and compared three types of models: a Decision Tree, Logistic Regression, and a Neural Network. The Decision Tree came out on top, correctly classifying outcomes about 88% of the time and it was slightly over the Neural Network (87.96%) and Logistic Regression (87.33%).

Based on these findings, a few practical suggestions can be made like concentrating policing efforts in high-crime areas, allocating more resources during the evening hours when arrest rates tend to be higher, and paying closer attention to districts where arrests remain persistently low.

1. Problem Description

1.1 Background and Description of the Problem

Crime stands as one of the most urgent public policy challenges for cities across the United States. Chicago, Illinois, consistently appears near the top of national rankings for both violent and property crime. For law enforcement agencies, city planners, and community organizations, understanding the dynamics of crime when it happens, where it happens, and what factors influence whether someone gets arrested or not is fundamental to build safer communities. As Ratcliffe (2010) points out, crime doesn't occur randomly; it tends to cluster in predictable geographic and time-based hotspots, which is exactly the kind of pattern that data mining is well-equipped to detect. Knowing where those hotspots are, and being able to anticipate arrest outcomes, gives police departments a real advantage when it comes to deploying limited resources strategically (Gorr & Harries, 2003).

The Chicago Data Portal offers one of the most detailed publicly available urban crime datasets in the country, with incident-level records going all the way back to 2001. This project focuses in on the years 2018 to 2020. This is a window that's particularly compelling because in this window there was arrival and early spread of the COVID-19 pandemic. This was a period when crime patterns shifted in documented and significant ways across the nation (Campedelli et al., 2020). Each record in the dataset includes details like crime type, geographic coordinates, time of occurrence, community area, district, ward, and whether an arrest was made. That combination of variables creates a rich, multi-dimensional picture well-suited to both clustering and predictive modeling.

1.2 Importance of the Problem

Being able to predict whether a crime will lead to an arrest has real, practical value for law enforcement policy. Over the past decade, predictive analytics in policing has grown considerably, with research showing that machine learning and data mining can genuinely improve how police departments operate (Perry et al., 2013). When arrest rates are consistently low for certain crime types or in particular neighborhoods, it often points to underlying gaps. Whether in enforcement capacity, evidence collection, or community trust. Identifying the factors that are close to arrest outcomes can help departments direct investigative resources more effectively and develop targeted intervention programs.

Clustering analysis for identifying crime hotspots fits naturally within the broader framework of evidence-based policing and geographic profiling (Weisburd et al., 2016). Pinpointing stable spatial patterns of crime can inform decisions about where to deploy patrols, how to structure community outreach, and where to invest in crime prevention infrastructure. Given the well-documented differences in crime distribution across Chicago's 77 community areas, this kind of analysis also carries meaningful implications for equity and fairness in public safety efforts.

1.3 Research Questions

Predictive Analysis Research Question: Can we predict whether a reported crime incident in Chicago (2018-2020) will result in an arrest based on crime type, geographic location, and time of occurrence?

Exploratory Analysis Research Question: Can crime incidents in Chicago be grouped into meaningful clusters based on spatial (latitude/longitude) and temporal characteristics to identify crime hotspots and behavioral patterns?

2. Data Exploration, Preparation, and Visualization

2.1 Data Set Description and Source

The dataset used in this project is the "Crimes - 2001 to Present" dataset published by the City of Chicago on the Chicago Data Portal. The dataset is publicly accessible at: https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2/about_data. The full dataset contains approximately 9.5 million records over two decades. For this analysis, records were filtered to the years 2018-2020, resulting in approximately 750,000 records after data cleaning.

Data Dictionary - Key Variables Used in Analysis:

Variable	Type	Description
Arrest	Binary (0/1)	Target variable - whether an arrest was made (1) or not (0)
Primary_Type	Nominal	Category of crime (e.g., Theft, Battery, Narcotics) - Input in predictive models
Location_Description	Nominal	Location type where crime occurred (e.g., Street, Residence) - Input in predictive models
Beat	Nominal	Chicago PD beat number - Input in predictive models
Community_Area	Nominal	One of 77 Chicago community areas - Input in both clustering and predictive models
District	Nominal	Chicago Police Department district number - Input in both analyses
Ward	Nominal	Chicago city ward number - Input in both analyses
Domestic	Binary (0/1)	Whether the crime was a domestic incident - Input in both analyses
Hour	Interval	Hour of the day (0-23) when the crime was reported - Input in both analyses
Month	Interval	Month of occurrence (1-12) - Input in both analyses
Year	Interval	Year of occurrence (2018-2020) - Rejected in both analyses
Latitude	Interval	Geographic latitude - Input in both analyses
Longitude	Interval	Geographic longitude - Input in both analyses

2.2 Data Cleaning, Preparation, and Modification

All data preprocessing was completed in Python before importing the data into SAS Enterprise Miner. The raw dataset which was downloaded from the Chicago Data Portal, covered incident-level crime records from 2001-present in CSV format. We applied the following preparation steps:

Year Filtering: We filtered the dataset to include only records from 2018 - 2020, bringing the total down from roughly 9.5 million rows to approximately 750,000 records which was filtered down by sample node to 187,500 records.

Date Parsing: We used Python's pandas library to parse and standardize the dates into proper datetime objects as the raw date column came had string formats. From there, we extracted Hour, Month, and Year as separate numeric columns.

Missing Value: Around 2% of records were missing geographic coordinates (Latitude and Longitude). If we had imputed spatial data then it would have introduced meaningful positional error and since 98% of records remained intact we chose to drop these rows. This is consistent with standard practice when missing geographic data cannot be reliably estimated (Han et al., 2011).

Boolean Conversion: The Arrest and Domestic columns were originally stored as True/False boolean values. We converted these to binary integers (1/0) to ensure compatibility with SAS Enterprise Miner's modeling nodes.

Variable Rejection: High-cardinality variables such as Description, ID and Year were excluded from modeling due to limited predictive value and excessive dimensionality. However, variables such as Primary_Type and Location_Description were retained because they provide strong predictive signals for arrest outcomes.

Sampling: Storage limitations in the UIS Citrix environment (1 GB cap) made working with the full dataset impractical. So, we used SAS Enterprise Miner's Sample node to draw a stratified random 25% sample. This produced approximately 187,500 records for clustering and a similar count for predictive modeling. For the predictive diagrams, data was partitioned 60/20/20 across training, validation, and test sets.

2.3 Data Visualization

To understand the distribution and spatial characteristics of the crime data before stepping into modeling, we developed several visualizations using Tableau Desktop Public Edition. Two key visualizations are presented below.

Figure 1. Crime incidents and arrest patterns by hour in Chicago (2018-2020).

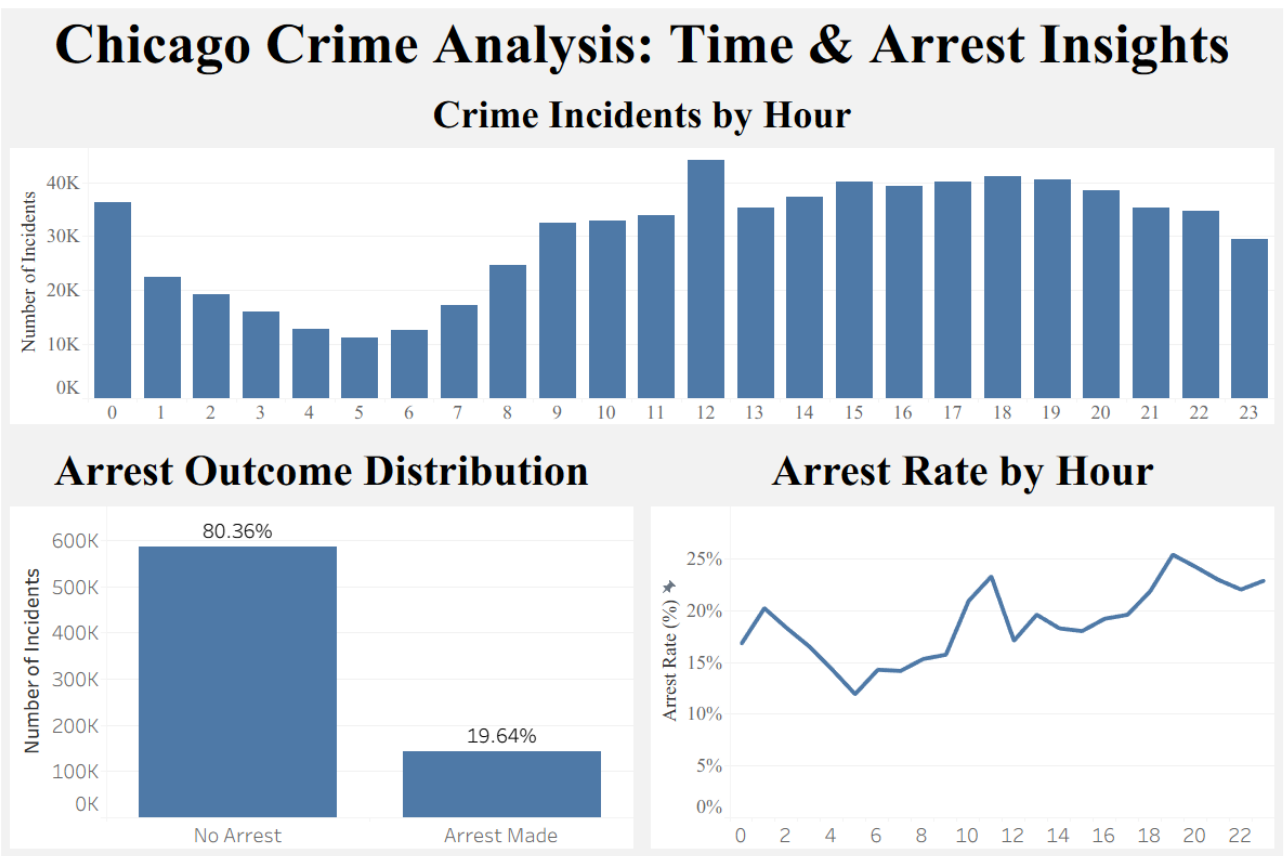

Figure 1 presents temporal crime patterns and arrest outcomes in Chicago. The bar chart shows the distribution of crime incidents by hour, revealing higher incident counts during afternoon and evening periods. The arrest outcome distribution indicates that approximately 80% of reported crimes did not result in an arrest, while roughly 20% resulted in arrests. The arrest rate by hour visualization highlights temporal variation in arrest probability, with higher arrest rates occurring during late evening hours. These temporal patterns suggest that time of day plays an important role in crime occurrence and arrest outcomes and therefore was included as a key predictor variable in the predictive modeling stage.

Interactive Dashboard:

An interactive version of the visualization dashboard is available at: **Chicago Crime Analysis: Time & Arrest Insights**

The dashboard allows users to interactively explore crime patterns by filtering year, hovering over hotspots, and examining temporal arrest patterns.

Figure 2. Geographic crime hotspot density and most frequent crime types in Chicago.

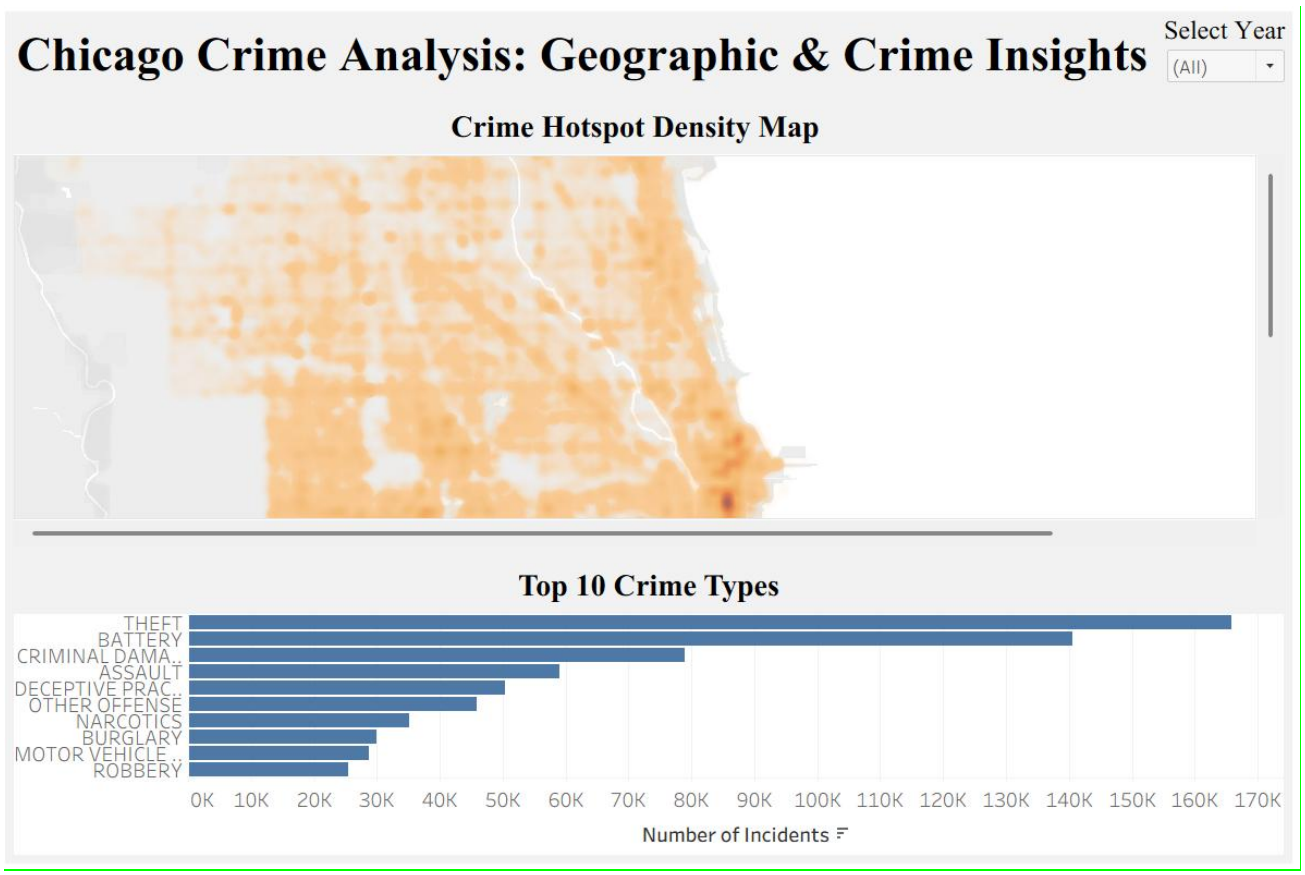

Figure 2 illustrates the spatial distribution of crime incidents across Chicago using a density map based on latitude and longitude coordinates. The visualization shows clear crime hotspots concentrated in central and southern regions of the city. The accompanying bar chart displays the ten most frequently reported crime types, with theft and battery representing the largest share of incidents. These spatial and categorical patterns confirm that crime in Chicago is not randomly distributed but instead clusters geographically and by crime category. This observation motivated the use of clustering techniques to identify stable spatial crime patterns.

Interactive Dashboard:

An interactive version of the visualization dashboard is available at: **Chicago Crime Analysis: Geographic & Crime Insights**

The dashboard allows users to interactively explore crime patterns by filtering year, hovering over hotspots, and examining temporal arrest patterns.

3. Exploratory Analysis (Clustering)

3.1 Overview and Methodology

To explore the underlying structure of Chicago's crime data, we applied unsupervised hierarchical clustering techniques in SAS Enterprise Miner. The goal was to determine whether crime incidents from 2018 to 2020 could be grouped into meaningful, stable clusters based on where and when they occurred. Unlike predictive modeling, clustering doesn't require a predefined outcome variable. It works by finding hidden patterns in the data on its own, which makes it well-suited for large-scale urban crime datasets (Han et al., 2011). We evaluated three hierarchical clustering methods: Ward, Centroid, and Average Clustering.

The clustering analysis was done on a 25% stratified sample of the cleaned dataset. The input variables included Arrest, Community Area, District, Domestic, Hour, Latitude, Longitude, Month, and Ward. We applied internal standardization and the specification method was set to Automatic, with a maximum of 10 clusters considered, a CCC cutoff of 3, a preliminary maximum of 50, and a final maximum of 20 candidate cluster solutions. To check how stable the resulting clusters were, we re-ran all three methods on a 90% sub-sample and compared the cluster structures to see how well they held up.

3.2 Clustering Results

Ward Clustering

The Ward method works by minimizing within-cluster variance at each merge step, making it well-known for producing compact, well-separated clusters (Everitt et al., 2011). For this reason, we designated it as our primary clustering approach. Based on the Cubic Clustering Criterion (CCC), SAS EM automatically identified three clusters as the optimal solution, with a clustering criterion value of 0.22191 and an incremental improvement of 0.007287 per iteration. The three clusters differed considerably in size: Segment 1 served as the cluster seed with just 6 observations, while Segment 2 contained 105,169 observations and Segment 3 contained 76,811. The Root-Mean-Square Standard Deviation (RMSSTD) values for Segments 2 and 3 were 0.222419 and 0.221203 respectively, suggesting similar levels of within-cluster compactness across both the groups. Cluster 2 was concentrated in specific community areas and districts on Chicago's south and west sides, while Cluster 3 showed a more spread-out geographic distribution across the city.

Centroid Clustering

The Centroid method merges whichever two clusters have the closest centroids at each step. In our 25% sample, it identified two clusters as optimal, with a clustering criterion value of 0.238082 and no measurable improvement between iterations. Segment 1 contained 181,980 observations, while Segment 2 held just 6 observations representing the centroid seed. The RMSSTD for Segment 1 was 0.238084, with a maximum distance from the cluster seed of 6.035938. Looking at the mean statistics, Cluster 1 had an average latitude of approximately 41.84 and Cluster 2 approximately 41.98.

Average Clustering

The Average method calculates the distance between two clusters as the average of all pairwise distances between their observation. This method produced results nearly identical to the Centroid solution where there were two clusters, the same clustering criterion value of 0.238082, and the same segment sizes (181,980 and 6 observations). The similar results of these two methods on the same structure shows that they're both picking up on the same fundamental pattern in the data.

3.3 Cluster Stability Analysis

To check how reliable our clustering solutions were, we applied all three methods to a 90% random sub-sample of the 25% training data and compared the results to what we found in the full sample. We performed this in order to check the stability of the clusters. It helps in confirming that the solution reflects genuine patterns in the data rather than quirks of a particular sample (Hennig, 2007).

Ward Clustering (90% Sub-Sample)

The Ward solution on the 90% sub-sample again produced three clusters, with a clustering criterion of 0.221913 which is nearly identical to the full-sample value of 0.22191. The improvement in clustering criterion was 0.00907, and the three segments contained 6 (seed), 69,205, and 94,576 observations respectively. The RMSSTD values (0.208753, 0.221303, and 0.222346) and maximum distances from the cluster seed (3.334632, 5.139769, and 5.695324) were closely in line with the full-sample results. Segment composition plots for Community Area and District showed consistent patterns across both samples, giving us confidence that the three-cluster Ward solution is robust and not just a byproduct of sampling variation.

Centroid and Average Clustering (90% Sub-Sample)

Both the Centroid and Average methods reproduced the same two-cluster solution on the 90% sub-sample, with clustering criterion values of 0.238081 and Segment 1 containing 163,781 observations alongside a 6-observation seed cluster. The near-perfect consistency between the full-sample and reduced-sample results shows that the two-cluster structure identified by these methods is highly stable and reliably reflects the underlying data.

3.4 Insights from Cluster Analysis

The clustering analysis revealed several meaningful patterns in how crime incidents were distributed across Chicago between 2018 and 2020.

Geography as the Primary Driver

Across all three clustering methods, incidents separated most strongly along geographic lines, with Latitude and Longitude emerging as the dominant differentiating variables. This reinforces the spatial patterns visible in the Hotspot Density Map and confirms that crime in Chicago is geographically concentrated enough to form well-defined and distinct clusters.

The Ward method's three-cluster solution gives us more actionable information than the two-cluster solutions produced by the Centroid and Average methods. The three clusters appear to correspond to: a small group of geographic outliers (Cluster 1), a large concentration of incidents in high-density South and West Side community areas (Cluster 2, ~105,000 observations), and a secondary cluster with a broader, more distributed geographic spread across the city (Cluster 3, ~77,000 observations).

Time of Day and Month as Secondary Factors

Hour and Month played a secondary role in differentiating the clusters, with some clusters showing notably higher average hours of occurrence. This suggests that temporal patterns layer on top of spatial ones which is a finding that is consistent with routine activity theory and which holds that crime tends to emerge from the convergence of motivated offenders, suitable targets, and a lack of capable guardianship at specific times and places (Cohen & Felson, 1979).

Community Area and District Patterns

Segment bar plots for Community Area and District showed clear compositional differences across clusters which confirmed that certain Chicago police districts and neighborhoods are tied to specific crime clusters. This has direct implications for designing geographically targeted crime prevention strategies and deploying resources where they're needed most.

4. Predictive Analysis

4.1 Modeling Framework and Data Partitioning

For the predictive phase of our analysis, we used three supervised data mining techniques: Decision Tree, Logistic Regression, and Neural Network. All models were built in SAS Enterprise Miner using the same 25% stratified sample of the cleaned dataset, with data split into Training (60%), Validation (20%), and Test (20%) sets via the Partition node. The target variable was Arrest which had a binary outcome where 1 indicates an arrest was made and 0 indicates it was not. Input variables included Beat, Community Area, District, Domestic, Hour, Latitude, Location Description, Longitude, Month, Primary Type, and Ward, while Description, ID, and Year were excluded from modeling.

We included Primary Type and Location Description intentionally as inputs here, even though they were rejected in the clustering diagram. The reason for this was that high-cardinality variables carry strong discriminatory signal for arrest outcomes. For example, narcotics incidents are associated with substantially higher arrest rates than theft, making crime category a genuinely useful predictor. To determine which model performed best, we used misclassification rate on the validation set as our primary evaluation criterion, which is consistent with established best practices for binary classification tasks (Breiman et al., 1984).

4.2 Decision Tree Analysis

Four decision tree configurations were evaluated, varying the maximum branch depth (B2 or B3) and maximum depth (D2, D4, or D6):

Model	Valid Misclassification	Train Misclassification	Accuracy (Valid.)
CrimeDecTree_B3D6 ✓ BEST	0.1159	0.1185	**~88.41%**
CrimeDecTree_B2D6	0.1197	0.1227	~88.03%
CrimeDecTree_B2D4	0.1234	0.1256	~87.66%
CrimeDecTree_B2D2	0.1286	0.1288	~87.14%

Table 2: Decision Tree Model Comparison Results

The best-performing decision tree was the CrimeDecTree_B3D6 configuration, which used 3 branches per split and allowed up to 6 levels of depth. This model achieved the lowest validation misclassification rate of 0.1159, with an accuracy of 88.41%, and was connected to the Score_FinalTree node for final scoring. The ROC curves across all four decision tree configurations, the B3D6 model consistently showed the largest area under the curve (AUC) on both the training and validation datasets, which confirms that it had the strongest overall ability to distinguish between arrest and non-arrest outcomes.

The cumulative lift chart further supported this selection. The B3D6 model achieved a cumulative lift of approximately 4.5 at the top 10% depth. Practically, this means the model is 4.5 times more likely to correctly identify arrest cases within the top-scored 10% of observations compared to random chance, which is a strong result for real-world operational use (Witten & Frank, 2005).

4.3 Logistic Regression Analysis

Four logistic regression models were estimated using different variable selection methods:

Model	Valid Misclassification	Train AIC	Accuracy
LR_Backward ✓ BEST	0.1267	71315.16	~87.33%
LR_Stepwise	0.1267	71315.16	~87.33%
LR_Exhaustive	0.1268	71399.65	~87.32%
LR_Forward	0.1268	71399.65	~87.32%

Table 3: Logistic Regression Model Comparison Results

The LR_Backward and LR_Stepwise models came out on top among the logistic regression configurations, both achieving a validation misclassification rate of 0.1267 (accuracy: 87.33%) and identical AIC values of 71,315.16. The LR_Exhaustive and LR_Forward configurations performed marginally worse, with slightly higher misclassification rates (0.1268) and AIC values of 71,399.65. We selected LR_Backward as the best logistic regression model.

The ROC curves for all four logistic regression models showed moderate and fairly consistent discriminatory ability, with the curves overlapping closely. This is a pattern that points to a shared limitation rather than meaningful differences between configurations. This probably shows that logistic regression uses a straight line to make decisions, which might not work well when the relationship between what we know and whether someone gets arrested isn't a straight line. Still, logistic regression is a useful tool here because we can easily understand how each predictor affects the chance of an arrest (Hosmer et al., 2013).

4.4 Neural Network Analysis

Four neural network configurations were evaluated, varying the number of hidden nodes (H3, H5, H7, H10):

Model	Valid Misclassification	Hidden Nodes	Accuracy (Valid.)
NN_H3 ✓ BEST	0.1204	3	~87.96%
NN_H10	0.1227	10	~87.73%
NN_H7	0.1256	7	~87.44%
NN_H5	0.1267	5	~87.33%

Table 4: Neural Network Model Comparison Results

Among the neural network configurations, the NN_H3 model using just 3 hidden nodes gave the best performance, achieving a validation misclassification rate of 0.1204 and an accuracy of 87.96%. This was somewhat surprising because it outperformed the more complex architectures: NN_H10 (87.73%), NN_H7 (87.44%), and NN_H5 (87.33%). It seems the simpler network did better, which might mean the bigger setups were a bit too focused on what they learned from the training data. This could be because they only used 25% of the data instead of the whole thing.

The ROC curves across all four neural network models showed strong and broadly comparable discriminatory ability on both the training and validation sets, with no single configuration standing out dramatically. The cumulative lift charts had a similar story. All neural network models achieved a lift of approximately 4.0 at the top 10% depth. While this is slightly below the Decision Tree B3D6 model's lift of 4.5, it still represents a strong improvement over random selection and reflects the neural network's solid overall predictive capability.

4.5 Overall Model Comparison and Scoring

The best-performing model from each technique was compared in the Overall Model Comparison node within SAS EM:

Model	Technique	Valid. Misclassification	Accuracy
CrimeDecTree_B3D6 ✓ OVERALL BEST	Decision Tree	0.1159	~88.41%
NN_H3	Neural Network	0.1204	~87.96%
LR_Backward	Logistic Regression	0.1267	~87.33%

Table 5: Overall Model Comparison - Best Model per Technique

The baseline misclassification rate of 11.35% corresponds to always predicting the majority class (No Arrest). Since approximately 88.65% of cases result in no arrest, a naive classifier would achieve about 88.65% accuracy without learning any patterns.

Our best Decision Tree model has a misclassification rate of 11.59%, which is very close to that baseline. So while the accuracy numbers look impressive on the surface, the model is not dramatically better than just always guessing "no arrest." This happens because arrests are relatively rare in the dataset only about 11% of crimes end in arrest which makes it hard for any model to stand out based on accuracy alone.

However, when we look at the ROC curve and cumulative lift chart, the model does show real value. The lift chart shows that if we take the top 10% of cases the model flags as likely arrests, we find 4.5 times more actual arrests than we would by random chance. This means the model is genuinely useful for prioritizing which cases to focus on, even if its overall accuracy does not look dramatically different from a simple guess. For datasets like this where one outcome is much rarer than the other, lift and ROC are better ways to judge model performance than accuracy alone.

To evaluate how well the model generalizes, we scored the CrimeDecTree_B3D6 on both the validation and test partitions using the Score node in SAS Enterprise Miner. The results were consistent across all three data partitions, giving us confidence that the model is not simply memorizing the training data but is capturing genuine patterns. The complete scored classification results are summarized in the table below:

Partition	Total Obs.	Predicted No Arrest (0)	Predicted Arrest (1)	Mean P(No Arrest)
Training	109,190	96,794 (88.65%)	12,396 (11.35%)	0.80
Validation	36,397	32,333 (88.83%)	4,064 (11.66%)	0.80
Test	36,399	32,242 (88.57%)	4,157 (11.43%)	0.80

Table 6: Score Node Classification Output - Train, Validation, and Test Partitions

The scoring results show several notable characteristics of how the model performs in practice. Most striking is the consistency of the predicted class distribution across all three partitions which is roughly 88.6-88.8% of observations are classified as no-arrest and 11.2-11.7% as arrest, closely reflecting the

actual class breakdown in the dataset. The mean predicted probability of no-arrest (P_Arrest0) holds steady at exactly 0.80 across the training, validation, and test sets, which is a strong indicator that the model is generalizing well rather than overfitting to the data it was trained on (Han et al., 2011).

When we take a closer look at the predicted probabilities, P_Arrest0 has a median of 0.87 and a 75th percentile of 0.96 across all partitions. This means the model is pretty confident that it won't arrest most of the observations. On the other hand, P_Arrest1 has a median of 0.13 but can go up to 1.00, indicating that the model can still spot some cases where an arrest is very likely. The consistent spread of predicted probabilities across all three partitions shows that the decision tree has learned decision boundaries that work well for new, unseen data. This is especially good news since the model was only trained on 25% of the full 2018-2020 dataset. It suggests that a good, representative sample was enough to capture the main patterns that affect arrest outcomes.

That being said, there's still room for growth. The class imbalance between arrest and non-arrest cases is still a challenge, and future research could look into methods like SMOTE oversampling for the smaller arrest class or using cost-sensitive modeling to make the model more aware of actual arrest cases (Chawla et al., 2002).

Although the improvement over the baseline accuracy is modest due to strong class imbalance, the Decision Tree model was selected as the final model because it achieved the lowest validation misclassification rate, demonstrated stable performance across partitions, and provided interpretable decision rules.

5. Conclusions and Practical Recommendations

5.1 How the Analysis Answers the Research Questions

Predictive Research Question

Our analysis shows that predicting arrest outcomes for Chicago crime incidents is not only possible but can be done with about 88% accuracy by using spatiotemporal and crime category features. The Decision Tree model (CrimeDecTree_B3D6) performed the best, showing that things like where the crime happened (community area, district, latitude, and longitude), when it happened (hour and month), and if it was a domestic incident are all important in predicting whether an arrest will occur. This directly answers our predictive research question: yes, arrest outcomes can be predicted with practical accuracy using the crime incident data from the Chicago Data Portal.

Exploratory Research Question

The clustering analysis shows that crime incidents in Chicago from 2018 to 2020 form some pretty clear and consistent patterns. The Ward linkage method found three stable groups, and both the Centroid and Average methods pointed to two. To make sure these clusters were real, we replicated the analysis with 90% of the data, and the Ward criterion values were pretty similar (0.22191 for the full sample and 0.221913 for the smaller sample). This small difference really backs up how solid the solution is. Overall, the clusters were mainly based on where things are geographically and what communities are like, which means we can use this spatial structure to create interventions that really hit the mark.

5.2 Practical (Actionable) Recommendations

Geographic Hotspot Policing

Both the Crime Hotspot Density Map and our clustering results point to the same conclusion that crime in Chicago is heavily concentrated in specific south and west side community areas. The Chicago PD should prioritize patrol density and rapid response resources in these identified hotspot clusters. Research on evidence-based hotspot policing has shown that this kind of targeted approach can reduce crime by 20-30% in focused areas without simply pushing criminal activity into neighboring zones (Weisburd et al., 2016).

Temporal Resource Deployment

The Arrest Rate by Hour analysis highlights two times when arrest activity is higher: late morning around 11 AM (~23%) and late evening between 7 PM and 11 PM. By scheduling more staff and focusing on investigations during these hours, law enforcement can make the most of the increased chances of successful arrests.

Domestic Crime Intervention Programs

The Domestic variable proved to be a meaningful predictor in our models which suggested that domestic incidents follow distinct arrest dynamics compared to other crime types. Alongside traditional law enforcement response, community-based domestic violence intervention programs and coordinated social services could help address the causes of domestic crime, particularly in the high-density clusters identified in our analysis.

Predictive Scoring for Case Prioritization

The Decision Tree model with scores has some real potential for making things run smoother in the field. For incidents that are predicted to lead to arrests in the top 10% of cases, we could send them straight to special investigative teams. On the other hand, cases with low predicted arrest probability might benefit more from different ways to resolve them, like community mediation or diversion programs. This kind of prioritization could really help us use our investigative resources more effectively.

Model Maintenance and Monitoring

It's important to remember that these models were trained on data from 2018 to 2020. This was a period that includes the significant disruption of the COVID-19 pandemic. As conditions continue to normalize, the models should be periodically re-evaluated using more recent data to ensure their predictions remain accurate and relevant over time (Gorr & Harries, 2003).

6. Team Member Contributions

The collaborative project folder link: **MIS576SP26.InsightOps**

Team Member	Significant Contributions
Atharva Chaskar	❖ Identified and selected the Chicago Crimes dataset from the City of Chicago Data Portal. ❖ Filtered the dataset to the 2018-2020 period and performed data cleaning and preparation using Python (pandas), including date parsing, missing value removal, boolean-to-integer conversion, and variable selection. ❖ Defined the predictive and exploratory research questions guiding the project. ❖ Performed clustering analysis and assisted in interpreting clustering results for exploratory insights. ❖ Contributed to documenting and organizing the analytical workflow.
Ajay Jayshree	❖ Conducted predictive modeling in SAS Enterprise Miner, including Decision Tree, Logistic Regression, and Neural Network models. ❖ Performed model comparison, identified the best-performing model, and executed scoring on validation and test datasets. ❖ Implemented clustering methods (Ward, Centroid, and Average linkage) in SAS Enterprise Miner and analyzed cluster stability. ❖ Developed all interactive visualizations in Tableau, including the Crime Hotspot Density Map, Crime Incidents by Hour, Arrest Outcome Distribution, and Arrest Rate by Hour dashboards. ❖ Contributed to interpretation of model results and visualization insights.

Both team members collaborated on reviewing results, discussing insights, and refining the final project presentation.

References

- Block, C. R., & Block, R. (1995). Space, place and crime: Hot spot areas and hot places of liquor-related crime. In J. E. Eck & D. Weisburd (Eds.), *Crime and place* (pp. 145-184). Criminal Justice Press.
- Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and regression trees*. Wadsworth & Brooks/Cole.
- Campedelli, G. M., Aziani, A., & Favarin, S. (2020). Exploring the immediate effects of COVID-19 containment policies on crime: An empirical analysis of the short-term aftermath in Los Angeles. *American Journal of Criminal Justice*, 46(1), 704-727. https://doi.org/10.1007/s12103-020-09578-6
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321-357.
- City of Chicago. (2024). *Crimes - 2001 to present* [Dataset]. Chicago Data Portal. https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2
- Cohen, L. E., & Felson, M. (1979). Social change and crime rate trends: A routine activity approach. *American Sociological Review*, 44(4), 588-608. https://doi.org/10.2307/2094589
- Everitt, B. S., Landau, S., Leese, M., & Stahl, D. (2011). *Cluster analysis* (5th ed.). John Wiley & Sons.
- Gorr, W. L., & Harries, R. (2003). Introduction to crime forecasting. *International Journal of Forecasting*, 19(4), 551-555. https://doi.org/10.1016/S0169-2070(03)00089-5
- Han, J., Kamber, M., & Pei, J. (2011). *Data mining: Concepts and techniques* (3rd ed.). Morgan Kaufmann.
- Hennig, C. (2007). Cluster-wise assessment of cluster stability. *Computational Statistics & Data Analysis*, 52(1), 258-271. https://doi.org/10.1016/j.csda.2006.11.025
- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). John Wiley & Sons.
- Perry, W. L., McInnis, B., Price, C. C., Smith, S. C., & Hollywood, J. S. (2013). *Predictive policing: The role of crime forecasting in law enforcement operations*. RAND Corporation.
- Ratcliffe, J. H. (2010). Crime mapping: Spatial and temporal challenges. In A. R. Piquero & D. Weisburd (Eds.), *Handbook of quantitative criminology* (pp. 5-24). Springer.
- Weisburd, D., Telep, C. W., Hinkle, J. C., & Eck, J. E. (2016). Is problem-oriented policing effective in reducing crime and disorder? Criminology & Public Policy, 9(1), 139-172. https://doi.org/10.1111/j.1745-9133.2010.00617.x
- Witten, I. H., & Frank, E. (2005). *Data mining: Practical machine learning tools and techniques* (2nd ed.). Morgan Kaufmann.